

Examination Scheme for EoSE for Semester-VI

CA – Continuous Assessment

EoSE – End of Semester Examination

Regular Students –

Type of Examination	Course Code and Nomenclature	Scheme of Examination	Duration of Examination	Maximum Marks	Minimum Marks
Theory	BTH-76T-651 HEALTHCARE BIOTECHNOLOGY AND ENTREPRENEURSHIP	CA	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	BTH-76P-652 HEALTHCARE BIOTECHNOLOGY AND ENTREPRENEURSHIP - PRACTICAL	CA	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks
Theory	BTH-76T-653 BIOETHICS AND IPR	CA	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	BTH-76P-654 BIOETHICS AND IPR - PRACTICAL	CA	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks
Practical	BTH-76P-655 DISSERTATION* (PRACTICAL)	CA	1 Hr	30 Marks	12 Marks
		EoSE	03 Hrs	120 Marks	48 Marks

The theory question paper will consist of **two** parts, **A & B**.

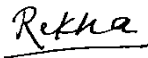
PART-A: 20 Marks

Part A will be compulsory having 10 very short answer-type questions (with a limit of 20 words) of two marks each.

PART-B: 60 Marks

Part B of the question paper shall be divided into four units comprising question numbers 2-5. There will be one question from each unit with an internal choice. Each question will carry 15 marks.

***Dissertation-** Based on the lab work practical exam will be of 120 marks distributed as below:-
(a) 100 marks viva voce and (b) 20 marks report of dissertation (evaluation by external examiner)

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UG0804 – Three/Four Year Bachelor of Science (Bio-Technology)								
				Bio-Technology 3rd YEAR SEM-VI	Credits			
#	L e v e l	S e m e s t e r	Type	Title	L	T	P	Total
1.	5	VI	MJR	UG0804 - BTH-76T-651 HEALTHCARE BIOTECHNOLOGY AND ENTREPRENEURSHIP	4	0	0	4
2.	5	VI	MJR	UG0804-BTH-76P-652 HEALTHCARE BIOTECHNOLOGY AND ENTREPRENEURSHIP - PRACTICAL	0	0	2	2
3.	5	VI	MJR	UG0804 - BTH-76T-653 BIOETHICS AND IPR	4	0	0	4
4.	5	VI	MJR	UG0804 - BTH-76P-654 BIOETHICS AND IPR - PRACTICAL	0	0	2	2
5.	5	VI	MJR	UG0804- BTH-76P-655 DISSERTATION-PRACTICAL	0	0	6	6

UG0804 - BTH-76T-651 HEALTHCARE BIOTECHNOLOGY AND ENTREPRENEURSHIP

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
VI	BTH-76T-651	HEALTHCARE BIOTECHNOLOGY AND ENTREPRENEURSHIP			7	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Introductory course of Biotechnology				

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Course Objectives	➤ Understand the principles and types of vaccines, including traditional and advanced generation vaccines. ➤ Comprehend various biotechnological tools for disease diagnosis, treatment, and drug designing. ➤ Explore the role of microorganisms in agriculture, the environment, industry, and fuel generation. ➤ Analyze the applications of biotechnology in medical, forensic, and industrial sectors. ➤ Develop an understanding of entrepreneurship principles, types, and processes with relevance to biotechnology. ➤ Cultivate awareness about ethical values, motivation, and innovation related to entrepreneurial ventures in the bio-sector.
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Detailed Syllabus

BTH-76T-651 HEALTHCARE BIOTECHNOLOGY AND ENTREPRENEURSHIP

Unit-I

Disease prevention: Generation of vaccine-First generation vaccines, Second generation vaccine (Cell Culture vaccine), Third generation vaccine (Recombinant and subunit and synthetic vaccines), Fourth generation vaccine (DNA vaccine), Medical Applications; Edible vaccines. Disease Diagnosis: identification of genes causing genetic disorders, DNA/RNA probes, and Commercial potential of diagnostics.

15 Lectures

Unit-II

Treatment and Drug Designing: Recombinant technology for disease treatment, Interferons, Oligonucleotides, Artificial tissues/organs. **Drug designing-** Drug delivery and targeting, Gene therapy, Autoimmunity, DNA fingerprinting in Forensic sciences.

15 Lectures

Unit-III

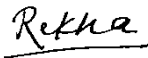
Entrepreneurship: Concept, Functions and need; Entrepreneurship: Characteristics and Theories; Relevance of Entrepreneurship to Socio-Economic Gain: Generating National Wealth, creating Wage and Self-Employment, Micro, Small and Medium Enterprises; Process of Entrepreneurship Development – Start-up and its stages.

15 Lectures

UNIT-IV

Types of Entrepreneurs: Competencies and characteristics: Ethical Entrepreneurship; Entrepreneurial Values, Attitudes and Motivation; Mindset of an employee and an entrepreneur - difference; Intrapreneur. **Innovations and Entrepreneurial Ventures** (Global and Indian) related to Biotechnology; New Industries of New Age Economy.

15 Lectures

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BTH-76P-652 HEALTHCARE BIOTECHNOLOGY AND ENTREPRENEURSHIP PRACTICAL

1. Mock design of first-, second, and third-generation vaccines using diagrams and flowcharts.
2. **ELISA (Enzyme-Linked Immunosorbent Assay)**-To detect antigens or antibodies as a model for disease diagnosis
3. Demonstration of DNA/RNA Probe Hybridization (Dot Blot or Southern/Northern Blot).
4. Case Study Analysis on Edible Vaccines
5. **DNA Fingerprinting using Agarose Gel Electrophoresis**- Simulate forensic applications and disease identification.
6. **PCR (Polymerase Chain Reaction) Demonstration or Virtual Lab**- Amplification of DNA for gene identification or disease-linked gene sequences.
7. **Study and Presentation on Artificial Organs and Interferons**- Literature-based group discussion or PowerPoint presentation.
8. Students form groups and develop a mini business plan related to a biotech product/service (e.g., diagnostic kits, organic fertilizers).
9. **SWOT Analysis Activity**- Identify Strengths, Weaknesses, Opportunities, and Threats for a biotechnology start-up.
10. **Case Study Analysis**- on successful biotech entrepreneurs or start-ups (e.g., Biocon, Serum Institute of India, Bharat Biotech).
11. **Mock Pitch Session**- Role-play: students act as entrepreneurs and present an idea to a panel of peers acting as investors.

Suggested Books & Text

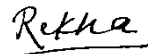
1. **Biotechnology: Expanding Horizons"** – B.D. Singh
2. **Introduction to Biotechnology"** – William J. Thieman, Michael A. Palladino
3. **Biotechnology Entrepreneurship: Starting, Managing, and Leading Biotech Companies"** – Craig D. Shimasaki
4. **Entrepreneurship Development"** – S.S. Khanka
5. **Innovation and Entrepreneurship"** – Peter F. Drucker
6. Fayolle A (2007) Entrepreneurship and new value creation. Cambridge, Cambridge University Press □
7. Hougard S. (2005) The business idea. Berlin, Springer
8. Lowe, R., & S Mariott (2006). Enterprise: Entrepreneurship & Innovation. Burlington, Butterworth Heinemann

E-learning resources:

1. https://onlinecourses.nptel.ac.in/noc20_mg35/preview
2. <https://kpu.pressbooks.pub/introtoentrepreneurship/>

Course Learning Outcomes:-

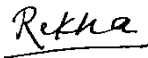
On successful completion of the course, students will be able to:

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1. **Explain** the classification and mechanisms of vaccine generations, including DNA and edible vaccines.
2. **Describe** modern tools for disease diagnosis, such as monoclonal antibodies, gene probes, and their commercial applications.
3. **Apply** recombinant DNA technology and gene therapy principles in the treatment of diseases and drug delivery systems.
4. **Analyze** the roles of microorganisms in bioremediation, agriculture, industrial product synthesis, and fuel generation.
5. **Assess** biotechnological innovations in industries including textile, paper, and surface deterioration control.
6. **Differentiate** between types of entrepreneurs and intrapreneurs, and explain entrepreneurial mindset, values, and motivation.
7. **Demonstrate** knowledge of entrepreneurship development processes, including start-ups and MSMEs.
8. **Evaluate** global and Indian entrepreneurial ventures and innovations in the biotechnology domain.

BTH-76T-653 BIOETHICS AND IPR

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
VI	BTH-76T-653	Bioethics and IPR			7	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Basic Knowledge of Introductory/Foundation level				
Course Objectives		➤ To understand the fundamentals of Intellectual Property Rights (IPRs) and their various forms in the context of biotechnology and agriculture.				

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	➤ To analyze national and international legal frameworks like TRIPS, WTO, WIPO, and the Budapest Treaty related to patents and plant breeders' rights. ➤ To examine issues related to patentability, commercialization, and biopiracy in biotechnology and traditional knowledge. ➤ To recognize ethical concerns and regulatory aspects of molecular biotechnology and genetically modified organisms (GMOs). ➤ To gain insights into the protection of traditional knowledge and the socio-economic and ecological implications of IPR and biosafety regulations.
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Detailed Syllabus BTH-76T-653 BIOETHICS AND IPR

Unit-I

Intellectual property rights: General Introduction to intellectual property rights, its different forms: Patents, trademarks, copyrights, trade secrets, Geographical indicators. Basic requirements of patentability, patentable subject matter, novelty, and the Public Domain; Nonobviousness, Patent infringements and revocation.

15 Lectures

Unit-II

Patentability of Biotechnology inventions and their commercialization, Budapest treaty. GATT, TRIPS, WIPO, Plant breeders' rights, international conventions, biopiracy. Biotechnology: Basic features of Indian Plant Varieties Protection & Farmer's Rights Act.

15 Lectures

Unit-III

Traditional Knowledge and Intellectual Property Rights: The importance of Traditional Knowledge (TK) for developing nations, protecting TK, The local, national and global dimensions of the issues in TK and IPRs, Traditional Medicine & IP Protection, Folklore, Patenting of Health Foods: Case studies

15 Lectures

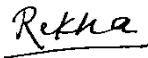
Unit-IV

Bioethics – Necessity of Bioethics, different paradigms of Bioethics – National & International. Ethical issues against the molecular technologies Regulatory Aspects: Direct Non-target effects on beneficial and native organisms, Indirect effects, Regulating Recombinant DNA technology, Biohazards, Environmental and regulatory aspects of using GM plants, Monitoring of introduced microorganisms, Biological weapons, Risk assessment. Brief account of bioethics in Biotechnology.

15 Lectures

BTH-76P-654 BIOETHICS AND IPR Practical

1. Proxy filing of an Indian Product patent

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2. Proxy filing of Indian Process patent
3. Planning for establishing a hypothetical biotechnology industry in India
4. A case study on clinical trials of drugs in India with emphasis on ethical issues.
5. Case study on women's health ethics
6. Case study on medical errors and negligence
7. Explain the different Forms for the grant of a patent
8. Comment on biopiracy with examples
9. Comment on biological weapons with examples
10. TRIPS, WTO, WIPO case studies

Suggested Books & Text

1. IPR, biosafety and Bioethics – Deepa Goel and Shomini Parashar
2. Bioethics and Biosafety – M K Sateesh
3. Plant biotech – Slater and Scott
4. Fleming, D.A., Hunt, D.L. (2000). Biotechnology and Safety Assessment (3rd Ed). Academic Press. ISBN-1555811804, 9781555811808.
5. Thomas, J.A., Fuch, R.L. (1999). Biotechnology and safety assessment (3rd Ed). CRC Press, Washington. ISBN: 1560327219, 9781560327219
6. Law and Strategy of biotechnological patents by Sibley. Butterworth publication.(2007) ISBN: 075069440, 9780750694445
7. H.K.Das. Textbook of Biotechnology, 3rd edition
8. Sateesh, M.K., Bioethics and Biosafety, IK International Publishers (2008)
9. Singh, I., and Kaur, B., Patent law and Entrepreneurship, Kalyani Publishers (2006)
10. Srinivasan, K. and Awasthi, H.K., Law of Patents, Jain Book Agency (1997)
11. Deepa Goel, Shomini Parashar (2013), IPR, Biosafety and Bioethics, Pearson.

E-learning resources

1. <https://archive.org/details/bioethicsprincip0000vaug/page/n5/mode/2up>
2. <http://rguir.inflibnet.ac.in:8080/jspui/bitstream/123456789/16305/1/9789354968761.PDF>
3. <https://www.icsi.edu/media/webmodules/publications/9.4%20Intellectual%20Property%20Rights.pdf>
4. <https://www.icsi.edu/media/webmodules/publications/9.4%20Intellectual%20Property%20Rights.pdf>

Course Learning Outcomes:

On successful completion of this course, students will be able to:

1. **Define and differentiate** various forms of intellectual property rights, including patents, plant breeders' rights, and farmers' rights.
2. **Explain** the development of patent systems in India and globally, including TRIPS, WTO, GATT, WIPO, and Budapest Treaty.
3. **Analyze** the legal, ethical, and regulatory implications of patenting biotechnology inventions and protecting traditional knowledge.

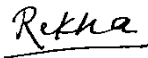
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4. **Evaluate** the relevance and application of laws such as UPOV, PVPFR Act, Bayh-Dole Act, and provisions like compulsory licensing and Bolar exemption.
5. **Discuss** major bioethical concerns related to GMOs, molecular biotechnology, and biological weapons.
6. **Identify** and **describe** biosafety levels, containment systems, and national guidelines for safe use of GMOs and LMOs.
7. **Illustrate** the role of regulatory bodies such as IBSC, RCGM, and GEAC in biosafety

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
VI	BTH-76P-655	Dissertation			7	6
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical (Presentation)	Total		
Intermediate	Major	-	6	6	NO	Lab work on a selected topic
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Intermediate course in Biotechnology				
Course Objectives		✓ To provide students with hands-on experience in designing, executing, and presenting independent research in any domain of biotechnology ✓ To enhance analytical and problem-solving skills through the application of experimental techniques in real-world biological systems. ✓ To develop proficiency in data collection, analysis, and interpretation using standard biotechnological tools and statistical methods. ✓ To train students in scientific communication skills, including the ability to write a structured dissertation report, present research findings, and defend conclusions logically and ethically.				

governance in India.

8. **Interpret** risk assessment, management, and communication strategies for biosafety in agricultural and medical biotechnology.

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BTH-76P-655 DISSERTATION

Report of Dissertation of 1-2 months: To improve career prospects in industry, clinical environments, in-house or other institute/training house/another university where state-of-the-art research facilities are available, is to be carried out on a selected topic. Students need to select a topic that has not been explored earlier and under the supervision of a faculty/trainer, have to carry out research in wet lab conditions. Students need to interpret results and prepare a report of dissertation at the end comprising, i) Introduction, ii) Review of literature, iii) Objectives of the present work, iv) Experimentation, v) Observation, and Results, vi) prospects of the work, (times new roman font size-12) vii) Figures and Tables (clear and original). The title page should comprise the Topic (font size 16), the name of the student, the name of the supervisor/trainer, name of the institute (font 14) with a diagrammatic representation of the work.

Evaluation Component	DISSERTATION REPORT (with proper certification)
Evaluation Method	Appropriate to the material and specific content
Learning Outcome	Designing and conducting research in a selected area of biotechnology, applying modern laboratory or computational techniques under guided supervision

Course Learning Outcomes:

Upon completion of the course, the student will know about

1. **Exposure to** hands-on experience in advanced instrumentation, experimental protocols, or bioinformatics tools relevant to academic, clinical, or industrial biotechnology.
2. **Critically analyze scientific data** and interpret results in the context of existing research, applying logical reasoning and evidence-based conclusions.
3. **Communicate scientific findings effectively**, both in written dissertation form and oral presentations, adhering to professional and ethical standards.
4. **Apply research-based learning** to address real-world problems, enhancing readiness for careers in biotechnology-related industries, clinical laboratories, or research institutions.

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